

Quantitative Research Methods

Multiple Testing

Advanced module A7 — optional self-study

Prof. Dr. Christoph Weisser

Slides, lab notebooks and data:  github.com/ChrisW09/Quantitative-Research-Methods

The advanced modules at a glance

Module	Topic
A1	Randomised Controlled Trials — potential outcomes, randomisation, power
A2	Explainable AI with Shapley Values — axioms, exact and Monte-Carlo Shapley
A3	Conformal Prediction — split conformal, CQR, prediction sets
A4	GLMs and Splines — exponential family, overdispersion, P-splines
A5	Support Vector Machines — margins, the cost C , kernels
A6	Survival Analysis — censoring, Kaplan–Meier, Cox
A7	Multiple Testing — FWER, Bonferroni and Holm, FDR
A8	Unsupervised Learning — PCA, K -means, hierarchical clustering
A9	Deep Learning — MLPs, convolutions, training, regularisation

Twelve optional, self-study modules sit outside the taught chapter plan (the table shows A1–A9; A10–A12 are newer additions — see the folder README); today's module is highlighted. Each is a full deck in the course house style with a companion lab notebook.

This module assumes Chapter 0 (hypothesis testing and p -values) and Chapter 5 (resampling). It was ISLP Chapter 13 in earlier editions of this course.

Contents

- 1 The Problem
- 2 FWER
- 3 FDR
- 4 Practice
- 5 Python Lab
- 6 Summary

Optional and advanced material is collected in the **appendix** at the end of the deck.

Notation in this chapter

Symbol	Meaning
m	number of hypotheses tested simultaneously
α	per-test significance level; target FWER
H_{0j}	j -th null hypothesis in the family
$p_j, p_{(j)}$	p -value of test j ; the j -th smallest (sorted) p -value
V	number of false positives (true nulls rejected)
R	total number of rejections, $R = V + S$
S	number of true positives (false nulls rejected)
U, T	true nulls retained; false nulls retained (missed effects)
FWER	family-wise error rate $\Pr(V \geq 1)$
FDR	false discovery rate $E[V / \max(R, 1)]$
q	target FDR level of Benjamini–Hochberg
Power	$E[S] / m_1$, with m_0 and m_1 the number of true and false nulls ($m_0 + m_1 = m$)

Sources and references

Primary textbook

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning, with Applications in Python*. Springer.

Learning objectives

By the end of this chapter you will be able to:

- **Explain** the multiple-testing problem and its consequences.
- **Apply** Bonferroni and Holm corrections to control family-wise error rate (FWER).
- **Apply** the Benjamini–Hochberg procedure to control false discovery rate (FDR).
- **Use** resampling-based p -values when the null distribution is hard to derive.
- **Avoid** the post-selection-inference and p -hacking pitfalls.

Roadmap of this chapter

Honest inference at scale

- 1 The multiple-testing problem.
- 2 FWER controls: Bonferroni, Holm.
- 3 FDR control: Benjamini–Hochberg.
- 4 Resampling-based p -values.
- 5 Practical issues: p -hacking, post-selection inference.

Outline

- 1 The Problem
- 2 FWER
- 3 FDR
- 4 Practice
- 5 Python Lab
- 6 Summary

Why naive testing fails at scale

Test m hypotheses simultaneously, each at level $\alpha = 0.05$.

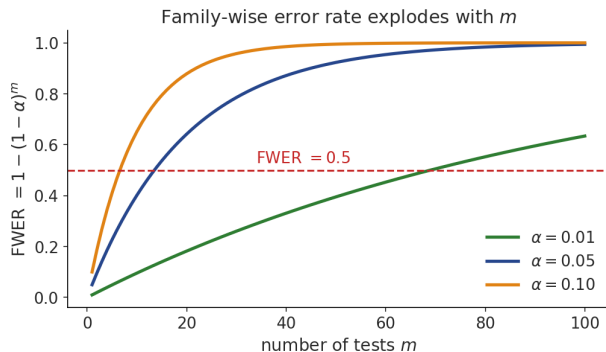
The arithmetic

- Even if all m nulls are true, we expect $0.05m$ false rejections.
- Test 10 000 genes $\Rightarrow \sim 500$ false discoveries by chance.
- Naive thresholding fails dramatically.

Why “some metric always wins”

Score one experiment on dozens of metrics and segments and this arithmetic guarantees that *something* comes back significant — every time, whether or not the change does anything. That is the mechanism by which a null change gets shipped to every user.

Why the family-wise error rate explodes



Takeaway

With $\alpha = 0.05$ the chance of ≥ 1 false positive passes 50% by only $m = 14$ tests and approaches certainty as m grows.

The classical hypothesis-test refresher

How to read this

- H_0 : null hypothesis. H_1 : alternative.
- Type I error: reject H_0 when true.
- Type II error: fail to reject when false.
- p -value: probability under H_0 of a test statistic at least as extreme as observed.
- “Reject H_0 if $p < \alpha$ ” controls the per-test Type I error rate at α .

Takeaway

A p -value is computed *assuming* H_0 is true; it is **not** the probability that H_0 is true, and a small p is not proof of a large effect.

The decision-table view

	Not rejected	Rejected	Total
H_0 true	U	V	m_0
H_0 false	T	S	m_1
Total	$m - R$	R	m

How to read this

- Family-wise error rate (FWER): $\Pr(V \geq 1)$.
- False discovery rate (FDR): $E[V/R \mid R > 0] \Pr(R > 0)$.
- Power: $E[S]/m_1$.
- Read it as an alert queue and the letters become staffing: R is the alerts raised, V those that waste an investigator's day, V/R the false-positive rate operations argues about, and T the fraud nobody ever sees.

Exercise 13.1 — The multiple-testing problem [Concept]

Exercise

- 1 Define a Type I error and a Type II error in terms of H_0 .
- 2 Suppose m hypotheses are tested independently, each at level $\alpha = 0.05$, and *all* nulls are in fact true. Write the probability of making at least one false rejection (the FWER) as a function of m .
- 3 Evaluate this probability for $m = 10$.
- 4 Find the smallest m for which the FWER reaches 0.5.

Solution — Exercise 13.1 (1/4)

Solution

Method. The FWER is easiest to reach through its complement: rather than count “at least one false positive” directly, find the single event “no false positive at all” and subtract from 1.

Part 1 — error types.

- Type I (false positive): rejecting H_0 when it is true.
- Type II (false negative): failing to reject H_0 when it is false.

Solution — Exercise 13.1 (2/4)

Solution

Part 2 — FWER formula. One test avoids a false positive with probability $1 - \alpha$. Under independence the m tests avoid it *jointly* with probability $(1 - \alpha)^m$ (independent probabilities multiply), so the complement is

$$\text{FWER} = \Pr(\text{at least one false positive}) = 1 - (1 - \alpha)^m.$$

Common mistake

Adding the per-test rates to get $m\alpha$ — that only approximates the FWER (the first-order term) and can exceed 1.

Solution — Exercise 13.1 (3/4)

Solution

Part 3 — $m = 10$.

$$1 - (0.95)^{10} = 1 - 0.5987 = 0.4013 \approx 40\%.$$

A 5% per-test rate has ballooned to a 40% chance of at least one spurious “discovery”.

Solution — Exercise 13.1 (4/4)

Solution

Part 4 — **FWER** = 0.5. Solve $1 - 0.95^m \geq 0.5$, i.e. $0.95^m \leq 0.5$:

$$m \geq \frac{\ln 0.5}{\ln 0.95} = \frac{-0.6931}{-0.05129} = 13.51,$$

so $m = 14$ tests already give a better-than-even chance of a false positive. This is exactly why corrections are needed.

Common mistake

Dividing by $\ln 0.95$ without flipping the inequality. Since $\ln 0.95 < 0$, $0.95^m \leq 0.5 \iff m \ln 0.95 \leq \ln 0.5 \iff m \geq \ln 0.5 / \ln 0.95$ — the sign flips at the last step.

Outline

- 1 The Problem
- 2 FWER
- 3 FDR
- 4 Practice
- 5 Python Lab
- 6 Summary

Bonferroni: split the error budget equally

Rule

Reject H_{0j} if $p_j \leq \alpha/m$.

How to read this

- m : number of tests in the family; α : target FWER (e.g. 0.05).
- Divide the budget evenly: each test is judged at the stricter level α/m .
- Equivalently, adjust each p -value to $\tilde{p}_j = \min(1, m p_j)$ and compare to α .

Takeaway

Controls FWER at α under *any* dependence. Conservative: it throws away power, but is simple, popular, and a good starting point.

Holm's step-down procedure (uniformly beats Bonferroni)

Algorithm

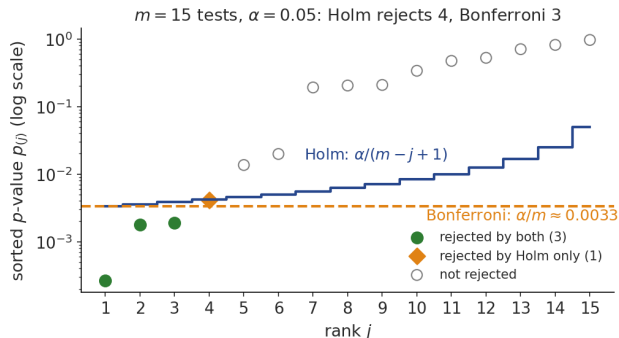
- 1 Order p -values $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$.
- 2 Find the smallest j with $p_{(j)} > \alpha / (m - j + 1)$.
- 3 Reject all $H_{0(k)}$ for $k < j$.

Why it is more powerful

- The thresholds *grow* down the list: $\frac{\alpha}{m}, \frac{\alpha}{m-1}, \dots, \alpha$ — the first equals Bonferroni, every later one is looser.
- “Step-down”: start at the smallest p -value and walk down, stopping at the first test that fails; keep all larger p -values.

Takeaway

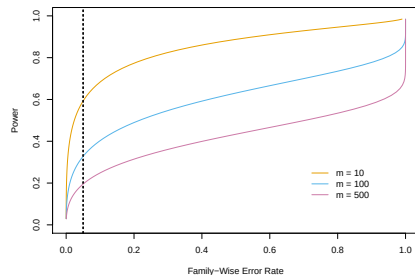
Uniformly more powerful than Bonferroni, yet still controls FWER at α .

Bonferroni vs. Holm on 15 simulated p -values

Takeaway

Bonferroni's flat bar $\alpha/m \approx 0.0033$ rejects 3 hypotheses; Holm's rising thresholds $\alpha/(m - j + 1)$ additionally catch $p_{(4)} = 0.0041$, rejecting 4 — Holm always rejects at least as many as Bonferroni.

Power loss with FWER control



Takeaway

Power collapses as m grows: at FWER = 0.05 (dashed) it falls from about 0.6 at $m = 10$ to about 0.2 at $m = 500$ — FWER control costs real effects.

Source: Figure 13.5 — ISLP, James et al. (2023).

Exercise 13.2 — Bonferroni correction [Math]

Exercise

You run $m = 6$ tests and obtain the sorted p -values

$$p = (0.001, 0.008, 0.012, 0.030, 0.040, 0.600),$$

and wish to control the FWER at $\alpha = 0.05$.

- 1 State the Bonferroni rejection threshold.
- 2 Which hypotheses are rejected?
- 3 Compute the Bonferroni-adjusted p -values $\tilde{p}_j = \min(1, m p_j)$ and confirm they give the same decisions.

Solution — Exercise 13.2

Solution

Step 1 — threshold. $\alpha/m = 0.05/6 = 0.00833$.

Step 2 — reject if $p_j \leq 0.00833$.

p_j	0.001	0.008	0.012	0.030	0.040	0.600
$\leq 0.00833?$	✓	✓	×	×	×	×

Reject H_1 and H_2 only ($0.008 < 0.00833$; $0.012 > 0.00833$).

Step 3 — adjusted p -values $\tilde{p}_j = \min(1, 6p_j)$:

(0.006, 0.048, 0.072, 0.180, 0.240, 1.000).

Rejecting when $\tilde{p}_j \leq 0.05$ again flags exactly H_1 (0.006) and H_2 (0.048). Two rejections.

Take-away. Bonferroni is simple and valid under any dependence, but conservative — here it misses H_3 at $p = 0.012$.

Common mistake: correcting twice — compare raw p_j with α/m , or adjusted $\tilde{p}_j$ with α . Comparing $\tilde{p}_j$ with α/m double-penalises.

Exercise 13.3 — Holm's step-down procedure [Math]

Exercise

Using the same data as Exercise 13.2,

$$p = (0.001, 0.008, 0.012, 0.030, 0.040, 0.600), \quad m = 6, \quad \alpha = 0.05,$$

apply Holm's step-down procedure. For each rank j compare $p_{(j)}$ with $\alpha/(m-j+1)$, stop at the first failure, and report the rejections. Compare with Bonferroni.

Solution — Exercise 13.3 (1/2)

Solution

The p -values are already sorted. Compare each with its Holm threshold $\alpha/(m - j + 1)$:

j	$p_{(j)}$	$\alpha/(m - j + 1)$	decision
1	0.001	$0.05/6 = 0.00833$	$0.001 \leq 0.00833 \checkmark$
2	0.008	$0.05/5 = 0.01000$	$0.008 \leq 0.01000 \checkmark$
3	0.012	$0.05/4 = 0.01250$	$0.012 \leq 0.01250 \checkmark$
4	0.030	$0.05/3 = 0.01667$	$0.030 > 0.01667 \times$

The first failure is at $j = 4$. Holm rejects all hypotheses with rank < 4 , i.e. $H_{(1)}, H_{(2)}, H_{(3)}$ — **three** rejections.

Solution — Exercise 13.3 (2/2)

Solution

Comparison. Holm also rejects $H_{(3)}$ at $p = 0.012$, which Bonferroni missed. Holm is uniformly more powerful than Bonferroni while still controlling the FWER at α , because its first threshold matches Bonferroni's but every later one is larger.

Common mistake: continuing down the list after the first failure. Holm is *step-down*: once $p_{(4)}$ fails, ranks 4–6 are all retained — even if a later $p_{(j)}$ were to sit below its own threshold.

Outline

- 1 The Problem
- 2 FWER
- 3 FDR**
- 4 Practice
- 5 Python Lab
- 6 Summary

The Benjamini–Hochberg (BH) procedure

Algorithm (control FDR at level q)

- 1 Order the p -values: $p_{(1)} \leq \dots \leq p_{(m)}$.
- 2 Find the largest L such that $p_{(L)} \leq Lq/m$.
- 3 Reject the L smallest hypotheses.

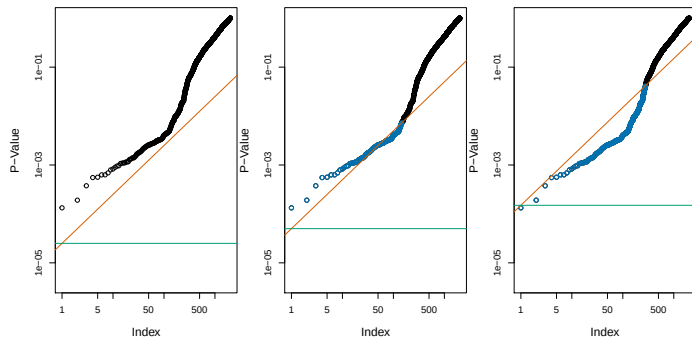
Reading the threshold Lq/m

- q : target FDR (e.g. 0.10); m : number of tests; L : rank in the sorted list.
- The bar Lq/m rises with rank, unlike Bonferroni's flat α/m .
- “Step-up”: scan from the *largest* p -value down; reject everything up to the first one that clears its bar.

Takeaway

Under independence (or positive dependence), BH controls FDR at q .

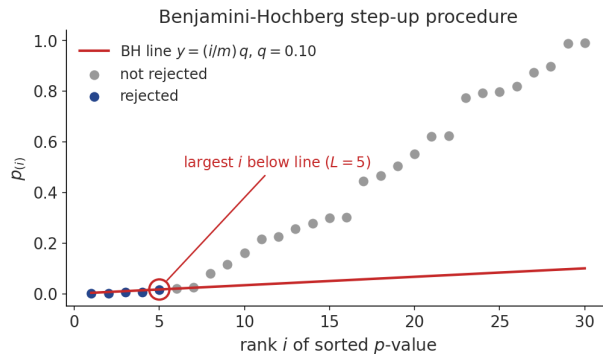
Visualising BH



Sorted p -values vs. the line Lq/m . Reject everything below the highest crossing.

Source: Figure 13.6 — ISLP, James et al. (2023).

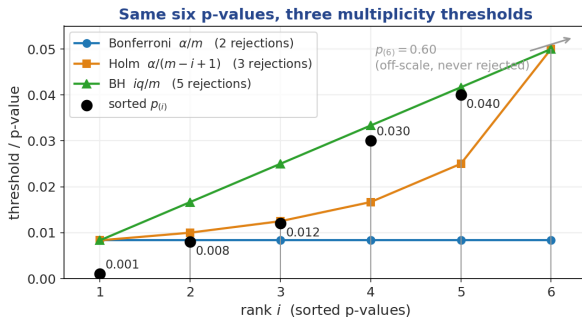
Benjamini–Hochberg as a staircase



Takeaway

Compare each sorted $p_{(i)}$ with the line $(i/m)q$; take the *largest* rank below it and reject everything up to there (5 of 30 here at $q = 0.10$).

Three thresholds on the same six p-values



Takeaway

A p -value is a candidate for rejection when it sits *below* its method's curve. All three meet at rank 1; then Bonferroni stays flat while Holm and BH rise — so here they reject 2, 3 and 5 hypotheses (Exercises 13.2/13.3/13.5).

FDR vs. FWER: which to use?

How to read this

- Use **FWER** when even one false discovery is unacceptable (drug-approval trials).
- Use **FDR** when an exploratory list of candidates is acceptable and false positives are tolerable in moderation (genomics, A/B testing).
- BH is much more powerful than Bonferroni when m is large and a non-trivial fraction of nulls are false.

Ask what the output is for

A false claim that is expensive and *public* — a drug label, a filing, a change shipped to every user — calls for FWER. A *ranked shortlist for further investigation* calls for FDR, because a known fraction of false leads is an acceptable cost of doing business. Where it is regulated the choice is not yours: FDA guidance on multiple endpoints requires a *pre-specified* multiplicity strategy.

Exercise 13.4 — FDR vs. FWER [Concept]

Exercise

- 1 Give the definitions of the FWER and the FDR in terms of the decision-table quantities V (false rejections) and R (total rejections).
- 2 Explain why controlling the FDR is generally less conservative (more powerful) than controlling the FWER.
- 3 For each scenario, state whether FWER or FDR control is more appropriate and why: (i) a confirmatory phase-III drug trial; (ii) screening 20 000 genes for follow-up experiments.

Solution — Exercise 13.4 (1/2)

Solution

Part 1 — definitions.

$$\text{FWER} = \Pr(V \geq 1), \quad \text{FDR} = E \left[\frac{V}{R} \mid R > 0 \right] \Pr(R > 0).$$

FWER is the probability of *any* false positive; FDR is the *expected proportion* of false positives among the rejections.

Part 2 — why FDR is more powerful. FWER guards against a single false positive anywhere, so its threshold must shrink with m (roughly α/m). FDR only requires the false positives to be a small *fraction* of discoveries, so it tolerates a few errors when there are many true discoveries. This lets it use a larger, adaptive threshold and reject far more when many nulls are false.

Solution — Exercise 13.4 (2/2)

Solution

Part 3 — scenarios.

- (i) **FWER** (Bonferroni/Holm): a single false approval is unacceptable; the cost of any Type I error is very high.
- (ii) **FDR** (Benjamini–Hochberg): an exploratory candidate list is the goal; a modest fraction of false leads is acceptable and will be filtered by follow-up experiments. FWER here would be far too conservative and kill power.

Common mistake

Defaulting to FWER “to be safe”. Safety is not free — the stricter criterion buys its guarantee with missed true effects, so the choice is a design decision, not a formality.

Exercise 13.5 — Benjamini–Hochberg procedure [Math]

Exercise

Using the same six p -values,

$$p = (0.001, 0.008, 0.012, 0.030, 0.040, 0.600), \quad m = 6,$$

apply the Benjamini–Hochberg (BH) procedure at FDR level $q = 0.05$.

- 1 For each rank L compute the BH threshold Lq/m .
- 2 Find the largest L with $p_{(L)} \leq Lq/m$.
- 3 State the rejections and compare with Bonferroni and Holm.

Solution — Exercise 13.5 (1/3)

Solution

Steps 1–2 — compare each $p_{(L)}$ with Lq/m , where $q/m = 0.05/6 = 0.00833$:

L	$p_{(L)}$	Lq/m	$p_{(L)} \leq Lq/m?$
1	0.001	0.00833	✓
2	0.008	0.01667	✓
3	0.012	0.02500	✓
4	0.030	0.03333	✓
5	0.040	0.04167	✓
6	0.600	0.05000	×

The largest L satisfying the condition is $L = 5$.

Solution — Exercise 13.5 (2/3)

Solution

Step 3 — reject. BH rejects the 5 smallest hypotheses $H_{(1)}, \dots, H_{(5)}$ (everything except $p = 0.600$).

Solution — Exercise 13.5 (3/3)

Solution

Comparison on identical data.

Bonferroni: 2 < Holm: 3 < BH: 5 rejections.

Controlling the FDR yields many more discoveries than controlling the FWER — at the price of allowing a small expected fraction ($\leq q$) of them to be false. Note BH is a *step-up* rule: even though $p_{(4)} = 0.030$ and $p_{(5)} = 0.040$ individually clear their thresholds here, BH would reject them via the largest passing L regardless of any earlier borderline case.

Common mistake

Treating q like α . They bound different things: α caps the chance of *any* false rejection; q caps only the expected *fraction* of the rejected list that is false.

Extended Exercise 13.1 — Full workflow on ten p-values [Math]

Extended exercise (15 min)

Ten hypotheses give the sorted p -values

$$p = (0.0008, 0.0041, 0.0060, 0.0100, 0.0180, 0.0290, 0.0850, 0.150, 0.240, 0.790),$$

with $m = 10$. Control the FWER at $\alpha = 0.05$ (Bonferroni, Holm) and the FDR at $q = 0.10$ (Benjamini–Hochberg).

- 1 List the rejections under **Bonferroni**, **Holm** and **BH**.
- 2 Verify the ordering $R_{\text{Bonf}} \leq R_{\text{Holm}} \leq R_{\text{BH}}$.
- 3 State the guaranteed FWER bound, and comment on what the FDR guarantee buys.

Solution — Extended Exercise 13.1 (1/3)

Solution — Bonferroni and Holm ($\text{FWER} \leq 0.05$)

Bonferroni: reject if $p_j \leq \alpha/m = 0.005$. Only $p_{(1)} = 0.0008$ and $p_{(2)} = 0.0041$ qualify $\Rightarrow$ 2 rejections. **Holm:** compare $p_{(j)}$ with $\alpha/(m - j + 1)$, stopping at the first failure:

j	$p_{(j)}$	$\alpha/(m - j + 1)$	
1	0.0008	$0.05/10 = 0.00500$	✓
2	0.0041	$0.05/9 = 0.00556$	✓
3	0.0060	$0.05/8 = 0.00625$	✓
4	0.0100	$0.05/7 = 0.00714$	× stop

Holm rejects ranks 1–3 $\Rightarrow$ **3 rejections**.

Solution — Extended Exercise 13.1 (2/3)

Solution — Benjamini–Hochberg ($\text{FDR} \leq 0.10$)

Compare each $p_{(L)}$ with $Lq/m = 0.01L$ and take the *largest* passing L :

L	$p_{(L)}$	Lq/m	
5	0.0180	0.050	✓
6	0.0290	0.060	✓ ← largest
7	0.0850	0.070	×
8	0.150	0.080	×

(Ranks 1–4 pass too.) The largest L with $p_{(L)} \leq Lq/m$ is $L = 6$, so BH rejects the six smallest $\Rightarrow$ **6 rejections** — including $p_{(6)} = 0.029$ and ranks 4, 5 that the FWER rules missed.

Solution — Extended Exercise 13.1 (3/3)

Solution — comparison and error guarantees

$$\underbrace{2}_{\text{Bonferroni}} \leq \underbrace{3}_{\text{Holm}} \leq \underbrace{6}_{\text{BH}}.$$

FWER bound. Bonferroni and Holm both guarantee $\text{FWER} = \Pr(V \geq 1) \leq \alpha = 0.05$ under *any* dependence: the chance of *even one* false rejection is at most 5%. **What FDR**

buys. BH instead guarantees $\text{FDR} = E[V/R] \leq q m_0/m \leq 0.10$. Among its 6 discoveries, at most an *expected* 10% ($\lesssim 1$ here) are false. Tolerating a small false *fraction* rather than *any* false positive is exactly why BH rejects more.

Common mistake

Reading “ $\text{FDR} \leq 0.10$ ” as “each discovery is 90% likely true”. The FDR is an average over the whole rejected list (and over repetitions), not a per-discovery probability.

Outline

- 1 The Problem
- 2 FWER
- 3 FDR
- 4 Practice**
- 5 Python Lab
- 6 Summary

p -hacking and selection bias

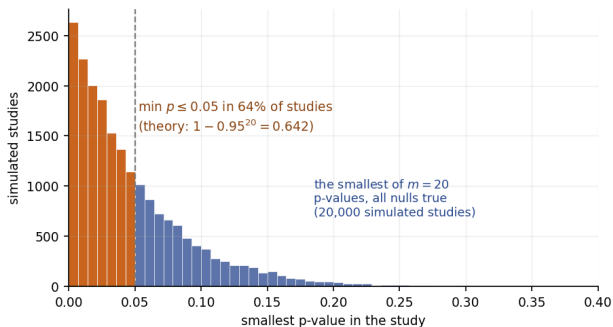
Don't

- Report only the most-significant subset of tests.
- Test thousands of hypotheses and call the smallest p -value “significant”.

Takeaway

Multiple-testing correction must include *every* test you ran, not only the ones you report. Pre-register hypotheses where possible; out-of-sample replication is the gold standard. Back-testing hundreds of strategy variants on one price history is the same failure wearing a different hat: an impressive Sharpe ratio arrives by chance, which is why practitioners deflate reported performance for the number of trials and demand out-of-time evidence before allocating capital.

What p -hacking harvests, drawn



Takeaway

Test $m = 20$ outcomes on pure noise and report the best: 64% of studies hand you a “significant” result ($1 - 0.95^{20} = 0.642$ — the same arithmetic as the FWER explosion, now read as a publication strategy). The smallest p -value in a fishing expedition is not a p -value.

Outline

- 1 The Problem
- 2 FWER
- 3 FDR
- 4 Practice
- 5 Python Lab**
- 6 Summary

Python lab — run it live in the notebook

Companion notebook: `advanced_07_multiple_testing_lab.ipynb`

The code in this section is meant to be **demonstrated live**. Open the companion Jupyter notebook and run it cell by cell:

- §1, *Simulation* — $m = 1000$ tests with a fraction of true alternatives;
- §2, *Compare procedures* — Bonferroni, Holm and Benjamini–Hochberg side by side;
- §3, *Visualise the Benjamini–Hochberg line*, then §4, *Permutation test*.

Data loads via the ISLP package or the bundled CSV files; §5, *Exercises* ends the notebook with hands-on tasks.

Bonferroni, Holm, BH in statsmodels

```
from statsmodels.stats.multitest import multipletests # corrections
import numpy as np # arrays

p = np.array([1e-6, 0.01, 0.04, 0.05, 0.30, 0.45, 0.80]) # m = 7 tests

for method in ["bonferroni", "holm", "fdr_bh"]: # FWER, FWER, FDR control
    rej, p_corr, *_ = multipletests(p, alpha=0.05, method=method) # adjust
    print(f"{method:12s}", rej) # True = reject Hj
    print(f"{'p_adj':12s}", np.round(p_corr, 4)) # compare with 0.05
```

How to read this

- **In:** the raw p -values of the *whole* family ($m = 7$ here) and the level $\alpha = 0.05$; `fdr_bh` reads it as the FDR target q .
- **Out:** `rej`, the reject flags, and `p_corr`, the *adjusted* p -values — pre-scaled (Bonferroni: $\min(1, mp_j)$) so you compare them with α itself, never with α/m .
- The printouts rank as the chapter predicts: Bonferroni fewest, Holm at least as many, BH most.

Extended Exercise 13.3 — Simulating FDR and power [Python]

Extended exercise (15 min)

Simulate $m = 1000$ one-sided z-tests of which $m_1 = 100$ have a true effect (mean shift $\mu = 3$); the other 900 are null. Over many repeats, apply Benjamini–Hochberg at $q = 0.10$ and Bonferroni at $\alpha = 0.05$, and estimate the *realised* FDR (V/R) and power (S/m_1) of each.

- 1 Write the simulation and run ~ 2000 repeats.
- 2 Report the realised FDR and power for BH and for Bonferroni.
- 3 Check BH against its theoretical bound $q m_0/m$, and comment on the power gap.

Run this live

The worked solution sits in `advanced_07_multiple_testing_lab.ipynb` under *Lecture exercises — worked Python solutions*: the cell headed *Extended Exercise 13.3 — Simulating FDR and power [Python]*.

Solution — Extended Exercise 13.3 (1/4)

Solution — code: setup and the BH rule

```
import numpy as np # arrays and random numbers
from scipy import stats # normal cdf for p-values
rng = np.random.default_rng(0) # seeded generator: reproducible
m, m1, mu = 1000, 100, 3.0 # 100 true effects, mean shift mu
q, alpha = 0.10, 0.05 # FDR target q, FWER target alpha
truth = np.arange(m) < m1 # ground truth: V and S need it

def bh_reject(p, q): # Benjamini--Hochberg step-up
    order = np.argsort(p); ps = p[order] # rank 1 = smallest p
    thr = (np.arange(1, m + 1) / m) * q # rising bar: (j/m) * q
    hit = np.where(ps <= thr)[0] # every rank under its bar
    rej = np.zeros(m, bool) # start: reject nothing
    if len(hit): rej[order[:hit.max() + 1]] = True # step up to largest
    return rej # mask in original order
```

Solution — Extended Exercise 13.3 (2/4)

Solution — code: simulate FDR and power

```
def simulate(method, reps=2000): # average FDR / power over reps
    fdr = pw = 0.0 # running totals
    for _ in range(reps): # one repeat = one study
        z = rng.normal(0, 1, m); z[:m1] = rng.normal(mu, 1, m1) # effects: mean mu
        p = 1 - stats.norm.cdf(z) # one-sided p-values
        rej = bh_reject(p, q) if method == "bh" else (p <= alpha / m) # BH or Bonf.
        V = (rej & ~truth).sum(); S = (rej & truth).sum(); R = rej.sum() # decision table
        fdr += V / max(R, 1); pw += S / m1 # FDP of this study; max avoids 0/0
    return fdr / reps, pw / reps # FDR = mean FDP over studies

for meth in ["bh", "bonferroni"]: # compare both procedures
    f, pw = simulate(meth) # run the experiment
    print("%-11s FDR=%.3f power=%.3f" % (meth, f, pw)) # read on the next slide
```

Solution — Extended Exercise 13.3 (3/4)

Solution — expected numbers and interpretation

Typical printout.

```
bh    FDR=0.090  power=0.723
bonferroni  FDR=0.002  power=0.186
```

- **BH** holds the realised FDR at 0.090 — essentially its theoretical ceiling $q_{m_0}/m = 0.10 \times 900/1000 = 0.09$ — while recovering about **72%** of the true effects.
- **Bonferroni** is far stricter: realised FDR ≈ 0.002 , but power collapses to about **19%** — it misses four-fifths of the signals.
- **Lesson:** at $m = 1000$, FDR control detects roughly *four times* as many true effects as FWER control, for a small, bounded false fraction.

Solution — Extended Exercise 13.3 (4/4)

Solution — expected numbers and interpretation

Common mistake: expecting the realised FDR to hit $q = 0.10$ exactly. The guarantee is $q m_0/m = 0.09$ here — BH adapts to the unknown share of true nulls, and any single run fluctuates around it.

Outline

- 1 The Problem
- 2 FWER
- 3 FDR
- 4 Practice
- 5 Python Lab
- 6 Summary**

Module A7 in one slide

What we accomplished

Honest inference when many hypotheses are tested at once.

- Set up the multiple-testing problem.
- Family-wise error rate (FWER): Bonferroni and Holm.
- False discovery rate (FDR): Benjamini–Hochberg.
- Resampling-based p -values and Storey's q -values.
- Pitfalls of post-selection inference and p -hacking.

Key formulas at a glance

FWER $\text{FWER} = \Pr(V \geq 1) = 1 - (1 - \alpha)^m$ (independent tests).

Bonferroni reject H_{0j} if $p_j \leq \alpha/m$ (controls FWER).

Holm reject while $p_{(j)} \leq \frac{\alpha}{m-j+1}$, stepping down from the smallest.

FDR $\text{FDR} = E\left[\frac{V}{\max(R, 1)}\right]$ (expected false-discovery proportion).

Benjamini–Hochberg reject the L smallest, where $L = \max\{j : p_{(j)} \leq \frac{j}{m}q\}$.

BH guarantee $\text{FDR} = E[V/R] \leq q$ $m_0/m \leq q$ (m_0 true nulls; independence or positive dependence).

Power $\text{Power} = E[S]/m_1$.

Vocabulary checklist

Term	Meaning
Type I error	Reject H_0 when true
Type II error	Fail to reject when false
$\text{FWER} = \Pr(V \geq 1)$	Family-wise error rate
$\text{FDR} = E[V/R]$	False discovery rate
Bonferroni / Holm	FWER controls
BH procedure	Step-up FDR control
Storey's q -value	FDR-calibrated p -value
Permutation test	Null distribution by relabelling
Knockoffs	Modern FDR control with covariate engineering

Decision rules of thumb

- Always specify the family of tests *before* looking at the data.
- Confirmatory / regulated $\Rightarrow$ FWER (Bonferroni or Holm).
- Exploratory / discovery $\Rightarrow$ BH on FDR.
- Report both raw and adjusted p -values.
- Replicate findings on an independent sample whenever possible.
- For valid post-selection inference: sample-splitting, selective inference, knockoffs.

Common pitfalls

Don't

- 1 Test thousands of hypotheses and report only the “significant” ones.
- 2 Use BH when tests are strongly negatively dependent.
- 3 Confuse FWER and FDR — they answer different questions.
- 4 Apply post-hoc multiple-testing fixes after looking at the data.
- 5 Use uncorrected p -values from stepwise / lasso.
- 6 Treat a p -value as a probability that H_0 is true.

Self-check questions

Quick self-assessment

- ① With m independent tests at $\alpha = 0.05$, what is the FWER — and at which m does it pass 50%?
- ② State Bonferroni's two equivalent formulations (threshold and adjusted p -value).
- ③ What is Holm's first threshold, and why does Holm never reject *less* than Bonferroni?
- ④ Is BH step-up or step-down, and what quantity does it control?
- ⑤ On the six p -values 0.001, 0.008, 0.012, 0.030, 0.040, 0.600 at $\alpha = 0.05$ (and $q = 0.05$): how many rejections under Bonferroni, Holm, BH?
- ⑥ You run 20 null tests and report only the smallest p -value. How often does it clear 0.05?

Answers

(1) $1 - 0.95^m$; $m = 14$. (2) Reject if $p_j \leq \alpha/m$, or if $\min(1, m p_j) \leq \alpha$ — the same rule. (3) α/m , the Bonferroni threshold; the later rungs $\alpha/(m-1), \dots$ only get easier, so Holm rejects everything Bonferroni does. (4) Step-up; the FDR, $E[V/R]$. (5) 2 / 3 / 5. (6) $1 - 0.95^{20} \approx 64\%$ of the time — reporting the winner is p -hacking.

Five things to remember

Module A7 in five lines

- 1 **Testing many things breaks α :** at $m = 14$ independent tests the FWER already passes 50%.
- 2 **Bonferroni is α/m under any dependence** — and Holm rejects at least as much for free, so use Holm.
- 3 **BH controls the FDR** with the rising thresholds $j q/m$ — the right guarantee for screening thousands of candidates.
- 4 **FWER for a few costly decisions; FDR for discovery at scale.**
- 5 **A p -value chosen after looking is not a p -value** — the smallest of 20 null tests clears 0.05 in 64% of studies.

What's next

Multiple testing is the module the rest of the material leans on:

- **Module A1 — Randomised Controlled Trials** treats peeking at a running experiment and scanning metric panels as exactly the multiplicity problem solved here.
- **Module A8 — Unsupervised Learning** ends where this module begins: clusters found in noise become claims only after corrected tests.
- The **final mock exam** and **Short Exam E** each carry a multiple-testing problem drawn from this module — Exercises 13.2, 13.3 and 13.5 are the drill.

Appendix — what is in here, and why

Nothing in the appendix is needed to follow the chapter: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

Contents of the appendix

- **Extended Exercise 13.2 — FWER or FDR? A design decision** — Exercise 13.4 expanded with arithmetic, over three solution slides. Run it from here if the concept version left doubts; otherwise it is homework.
- **The four outcomes, drawn** — a diagram of the same 2×2 table the main deck already gives as a table.
- **Resampling-based inference** — permutation p -values, the resampling estimate of the FDR, and what the histogram of many p -values looks like. Use it when the null distribution is not available in closed form.
- **Post-selection inference** — why a p -value computed after looking at the data is not a p -value. Important in research practice, beyond the exam.

Extended Exercise 13.2 — FWER or FDR? A design decision [Integrative]

Extended exercise (15 min)

For each study, choose an error criterion and justify it, using the naive expected number of false discoveries $E[V] = m_0 \alpha$ (all-null worst case $m_0 = m$) and the power consequences.

- 1 **Scenario A — confirmatory phase-III trial:** $m = 4$ co-primary endpoints; one false “significant” endpoint could license an ineffective drug.
- 2 **Scenario B — exploratory screen:** $m = 20\,000$ genes tested for follow-up; a shortlist with a few false leads is acceptable.
- 3 Contrast the statistical power the two criteria leave on the table.

Solution — Extended Exercise 13.2 (1/4)

Solution — scenario A: control FWER

With $m = 4$ tests at $\alpha = 0.05$, naive testing expects

$$E[V] = m_0\alpha = 4(0.05) = 0.20 \text{ false positives,} \quad \text{FWER} = 1 - 0.95^4 = 0.185.$$

A $\sim 19\%$ chance of *any* false endpoint is unacceptable when the decision is regulatory and irreversible. **Choose FWER control** (Bonferroni threshold $\alpha/m = 0.0125$, or Holm). One false positive is the outcome we must not allow, so we pay the power cost gladly — with only $m = 4$ tests that cost is tiny.

Solution — Extended Exercise 13.2 (2/4)

Solution — scenario B: control FDR

With $m = 20\,000$ and (worst case) $m_0 \approx 20\,000$, naive testing expects

$$E[V] = m_0\alpha = 20\,000(0.05) = 1000 \text{ false discoveries!}$$

Bonferroni would shrink the threshold to $\alpha/m = 2.5 \times 10^{-6}$, cutting $E[V]$ to ≈ 0.05 but **destroying power** for realistic effect sizes. **Choose FDR control** (BH at $q = 0.10$): accept that an expected $\leq 10\%$ of the shortlist is false in exchange for detecting far more true signals — the false leads are removed by follow-up experiments.

Solution — Extended Exercise 13.2 (3/4)

Solution — the power trade-off

	Confirmatory ($m = 4$)	Exploratory ($m = 20\,000$)
Criterion	FWER (Bonferroni/Holm)	FDR (BH)
Threshold shape	α/m (fixed, tiny)	adaptive Lq/m
Naive $E[V]$	0.20	1000
Priority	no false positive	many true findings

Solution — Extended Exercise 13.2 (4/4)

Solution — the power trade-off

Trade-off. The FWER threshold shrinks like α/m , so power collapses as m grows — fine for a handful of endpoints, ruinous for 20 000 genes. FDR asks only that false positives be a small *fraction* of discoveries, so its adaptive threshold keeps power high at scale. **Rule of thumb:** match the criterion to the cost of a single false positive.

Common mistake

Quoting $E[V] = m\alpha$ as a fact about your study — it is the all-null worst case ($m_0 = m$). With real effects present the count is smaller, though still ruinous at scale.

The four outcomes of a test decision

	Not rejected	Rejected	
H_0 true	U true negatives	V Type I (false pos.)	m_0
H_0 false	T Type II (false neg.)	S true positives	m_1
	$m - R$	$R = V + S$	

Takeaway

Type I errors are cell V — rejected true nulls. The $\text{FWER} = \Pr(V \geq 1)$ guards against *any* false positive; the $\text{FDR} = E[V/R]$ controls their *expected fraction* among the R rejections.

Resampling-based p -values

When the test statistic's null distribution is hard to derive:

How to read this

- **Permutation**: shuffle the response many times to get a null distribution.
- **Bootstrap (under H_0)**: resample under a constraint that enforces the null.
- Empirical p -values are then thresholded by Bonferroni / Holm / BH as before.

Takeaway

Empirical p -value = $\frac{1 + \#\{\text{resampled statistics} \geq \text{observed}\}}{1 + B}$ over B resamples — no parametric null needed. The “+1” keeps it strictly positive.

Resampling-based FDR

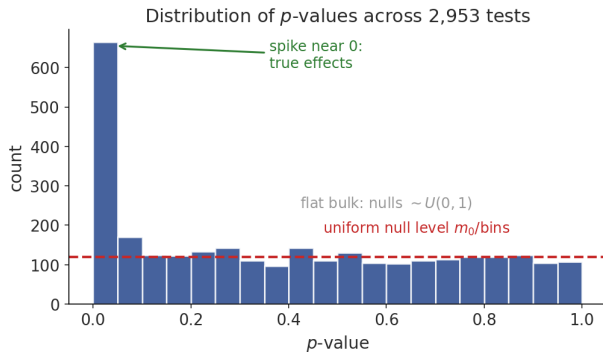
Storey's q -value

Estimate the proportion of true nulls $\hat{\pi}_0$ from the p -value distribution. Calibrate FDR more accurately than BH. Especially relevant when many nulls are false.

How $\hat{\pi}_0$ is read off

True nulls give p -values that are $\sim U(0, 1)$, so the histogram is *flat* away from 0; the height of that flat part estimates $\hat{\pi}_0$. BH implicitly assumes $\hat{\pi}_0 = 1$; plugging in the smaller true value makes FDR control less conservative and recovers more discoveries.

The distribution of p -values under many tests



Takeaway

True nulls give p -values that are $\sim U(0, 1)$ (the flat bulk), while true effects pile up near 0. Storey's method reads $\hat{\pi}_0$ off the height of that flat part to calibrate the FDR.

Post-selection inference

After variable selection (lasso, stepwise) standard p -values are invalid.

Takeaway

The bias: the *same data* both *chose* the hypothesis and *tested* it, so the reported p -values are optimistically small. Valid inference must account for the selection step.

Modern methods

- Sample splitting / data-carving.
- Selective inference (Lee, Sun, Sun, Taylor).
- Knockoffs (Barber, Candès).